



Smart Sensors For Sustainable Agriculture

Researchers from the University of Glasgow, Central South University of Forestry and Technology, and the Institute of Microelectronics and Photonics have developed compostable smart sensors to reduce e-waste in digital agriculture.



Soil health sensor (Credit: www.hackster.io)

These sensors feature a reusable electronic module (Microchip ATmega328P, Zigbee-enabled) and a biodegradable patch made from polymer substrate, graphene-carbon ink, and molybdenum disulfide. The patch, produced via screen-printing, decomposes naturally. Lab tests confirmed the system accurately monitored soil pH (3-8) and detected ethephon. Replacing only the patch every three months could cut environmental impact by up to 79% over five years.

Wearable Tech Simulates Realistic Touch

Engineers at Northwestern University (United States) have created a lightweight, wearable haptic device that mimics complex touch sensations like sliding, twisting,



This haptic device reproduces complex sensations (Credit: Northwestern University)

and pressing. Unlike typical devices that only vibrate, this one uses tiny multidirectional actuators with magnets and coils to simulate pressure, stretching, and motion. It connects via Bluetooth to phones

or VR headsets and includes an accelerometer to respond to user movement. The device can convert music into touch signals and may enhance VR, aid the visually or hearing impaired, and improve online shopping.

Clean Transfer Of 2D Semiconductors

Researchers from Peking University (Beijing, China) and the Beijing Graphene Institute (Beijing, China) have developed a wafer-scale method to integrate 2D semiconductors

with dielectric materials, significantly reducing defects during transfer. The team achieved a clean, polymer-free transfer using single-crystal antimony oxide (Sb_2O_3) deposited on graphene grown on Cu(111) via vacuum thermal evaporation. A water-ethanol pretreatment eased separation, while Sb_2O_3 also served as protective encapsulation. This enabled the defect-free transfer of a 10.16cm graphene wafer with high carrier mobility ($\sim 14,000 \text{ cm}^2\text{V}^{-1}\text{s}^{-1}$) and stability.

Simple Test To Check Battery Overheating

Researchers at the University of Tokyo have developed a safer, low-cost method to study lithium-ion battery failures. They created a mini test battery—just 1/50th the size of a regular one—that's easier to overheat, allowing safer thermal runaway research in small labs. The team also introduced the thermal runaway factor, a simple equation for overheating risk using battery heat capacity, energy, and surface area. By altering the battery's shape and removing non-heating parts, the design minimises heat release. This approach aids safer battery development and supports cleaner energy innovation.

Mass-Produced Cancer-Fighting Nanoparticles

MIT (USA) researchers have developed a scalable method to manufacture drug-loaded, polymer-coated nanoparticles using a new microfluidic mixing technique. This approach enables rapid, large-scale production—about 50 doses in minutes—compared to nearly an hour with traditional methods. The nanoparticles, created using layer-by-layer assembly, carry therapeutic drugs and targeting molecules that bind to tumor sites and release treatments locally. In mouse models, IL-12-loaded particles delayed tumor growth and showed potential for cures. Initially targeting ovarian cancer, the FDA-aligned method may extend to other cancers.

DNA To Build 3D Electronic Devices

Columbia Engineering (New York) researchers have used DNA origami to build 3D, nanoscale electronic devices, a leap from traditional 2D microchip design. This bottom-up method relies on DNA's natural base-pairing to self-assemble precise structures. Led by Aaron Michelson, the team anchored DNA to gold squares, forming octahedral frames. These were then coated with silicon oxide and tin oxide—essential semiconductor materials—and fitted with electrodes. The resulting devices function as light sensors, demonstrating that DNA-based 3D frameworks can be integrated into microchips, promising denser, brain-like architectures that are ideal for advanced AI and computing systems.